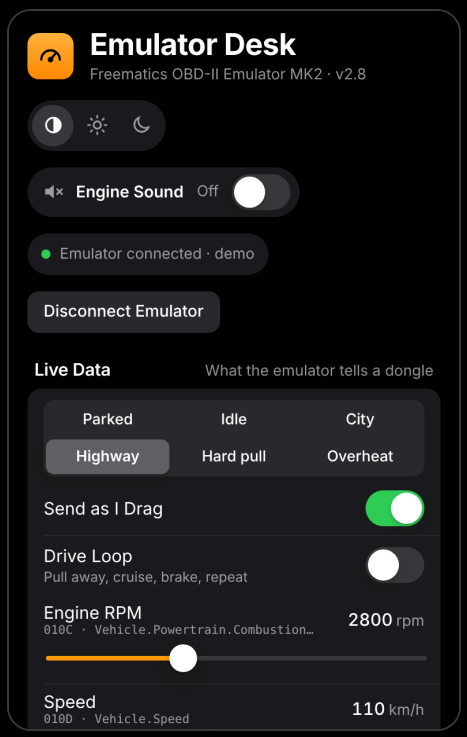
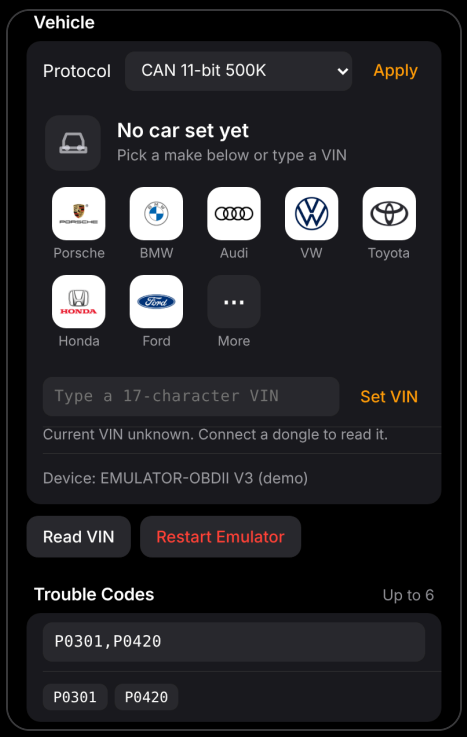
Emulator Desk

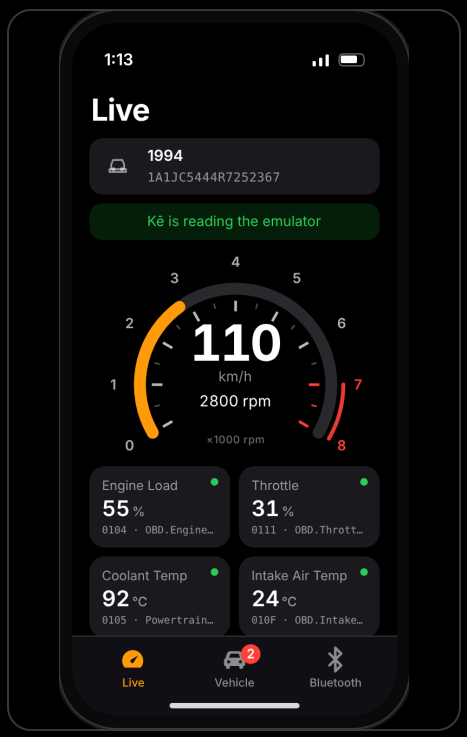
A Mac control panel for the Freematics OBD-II Emulator MK2

App **v2.8**
Document **v2.8**

Emulator Desk drives a Freematics MK2 emulator from Google Chrome over its USB cable, so a Mac can do what the Windows-only Freematics GUI does. You set live engine values, the VIN, the bus protocol and trouble codes, and any OBD-II dongle or scan tool plugged into the emulator reads them as if it were in a real car. It runs from a web page, with nothing to install, and follows Apple's Human Interface Guidelines in light and dark mode.







Live data

Ten sliders for the common Mode 01 values, with presets from Parked to Overheat. Values go out as you drag.

Vehicle and codes

Protocol, VIN and up to six trouble codes. Tap a car maker to load a sample VIN; the car and its logo show above.

Kē phone view

A phone with a tab bar. Live has an rpm gauge with redline, Vehicle decodes or scans a VIN, Bluetooth shows the link and what the dongle supports.

10
live PIDs with sliders

6
driving presets plus a drive loop

7
bus protocols to choose from

1
link to open it, nothing to install

On your phone

phone.html in Bluefy calls the dongle the Kē, connects from the Live tab, reconnects if it drops, keeps the screen on and can scan a VIN barcode.

Dongle over Bluetooth

Any ELM327 dongle shows as the Kē. It reads classic OBD-II, or OBD on UDS for 2027 cars and EVs, and says which on the Bluetooth tab.

Good to know

Sliders read back the emulator's values on connect. Its firmware can't report its VIN, so the VIN box fills in from the dongle.

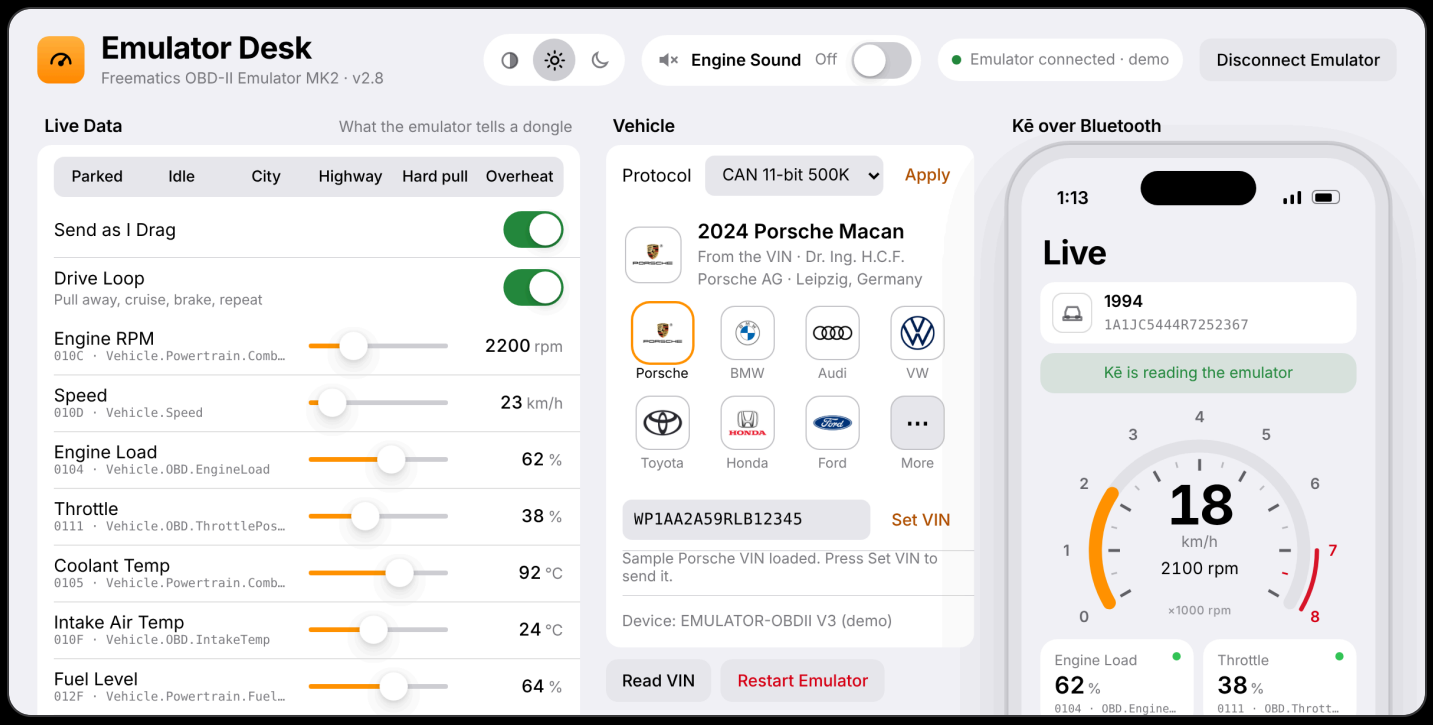
Engine sound

Flip the Engine Sound switch and a flat-six plays, built from single cylinder firings. It thumps at idle, howls near redline and crackles on lift-off.

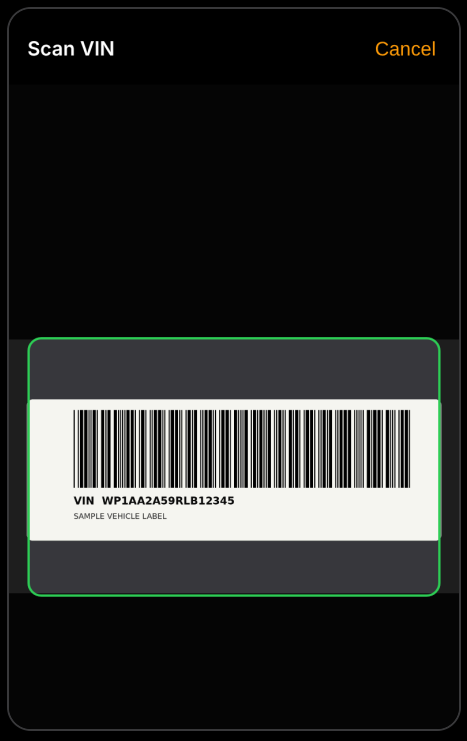
Emulator Desk

Appearance, car logos and the VIN scanner

App **v2.8**
Document **v2.8**



Light mode. The switch beside Engine Sound picks Auto, Light or Dark, and the choice is remembered. Here a sample Porsche VIN is loaded, so the Vehicle panel shows the car and its logo.



Scan VIN

The barcode button beside Decode opens the camera and reads the door-jamb or windshield barcode on its own.



Decoded car

Decoded straight away: logo, model, year and plant, then engine, body and drivetrain from NHTSA.

How scanning works

It reads Code 39, Code 128, Data Matrix, QR and PDF417, drops the leading "I" that door-jamb labels add, and prefers a VIN whose check digit is valid. A barcode that isn't a VIN gets a hint instead of a wrong result.

No live camera?

If the browser can't show the camera, or access is off, Take a Photo opens the camera app or photo library and the photo is read instead.

Car logos

Logos always sit on a white tile so they read in both modes. If one can't load, a plain car symbol shows.

Technical setup

Sources, integrations and what runs where

App **v2.8**
Document **v2.8**

Sources and libraries

Piece	Version	What it does	Loaded from	Licence
Emulator Desk	2.8	The whole app: UI, command queue, PID encoding, presets, drive loop, engine sound, dongle check, VIN decoder	`index.html` and `phone.html` on GitHub Pages	Written for this app
Freematics AT command set	EMULATOR-OBDII V3 (tested)	Commands the app sends: `ATE0`, `ATZ`, `ATSET`, `ATGET` (PIDs), `ATCLR DTC`	Built into the emulator	Freematics
OBD-II PIDs and VSS names	SAE J1979, J1979-2/-3 (UDS); COVESA VSS 4	Each value is tracked by its VSS name (e.g. Vehicle.Speed) and fetched by its PID, with byte encodings such as RPM × 4	Hard-coded in the app	Public standard
barcode-detector (with zxing-wasm)	3.2.3 (zxing-wasm 3.1.6)	Reads VIN barcodes where the browser has no built-in barcode reader (Safari, Bluefy)	jsDelivr CDN, loaded only when you first tap Scan	MIT; ZXing C++ Apache-2.0

No other libraries or web fonts; type is the system font.

Integrations

Integration	Used for	Details
Web Serial API	Talking to the emulator	`navigator.serial.requestPort()` filtered to CP2102 and CH340 USB chips, 38400 baud, 8N1. You pick the port in Chrome's chooser.
Freematics OBD-II Emulator MK2	The device controlled	USB serial through a CP2102 or CH340 chip, drivers built into macOS. Needs its 12V adapter.
Web Bluetooth	Reading the dongle	`navigator.bluetooth.requestDevice()`, you pick the dongle. Finds a notify and a write characteristic (FFE0, FFF0, 18F0 and similar) and sends ELM327 commands. Reads classic Mode 01 (`010C`), or OBD on UDS (`22F40C`, `22F802`, `1942`) when the car only answers that.
NHTSA vPIC	VIN details	`vpic.nhtsa.dot.gov/api/vehicles/DecodeVinValues/{VIN}?format=json`, no key. Sends only the VIN. Maker, country, year and check digit decode offline.
Camera and BarcodeDetector	Scanning a VIN barcode	`getUserMedia` with the rear camera; frames go to `BarcodeDetector` or zxing-wasm every 180 ms, with a photo picker as fallback. Images never leave the phone.
Asetrix logos API	Car maker logos	`api.asetrix.online/api/v1/logos/{make}`, free, no key. Sends only the make name, such as "porsche"; returns a PNG.
Web Audio API	Engine sound	An AudioWorklet flat-six model with exhaust resonators, muffler filter and compressor. Starts only when you flip the switch.
GitHub Pages	Hosting	`davidanthonyalleyne-dot.github.io/emulator-desk`, served from the `main` branch of the public `emulator-desk` repo. Static files only.

What runs where

On the device: everything. The page runs in Chrome or Bluefy and sends one emulator command at a time.

Network: the page itself, NHTSA per new VIN, a logo per make, and the barcode reader on first scan.

Data kept: two browser settings, `edTheme` (appearance) and `ed.lastDongle` (phone). Values, VIN and codes stay in the page and on the emulator.

Demo mode: add `?demo` to try it without hardware. The screenshots use it with a sample VIN and label. `node shoot.js` retakes them; `node pdf.js` prints this PDF.